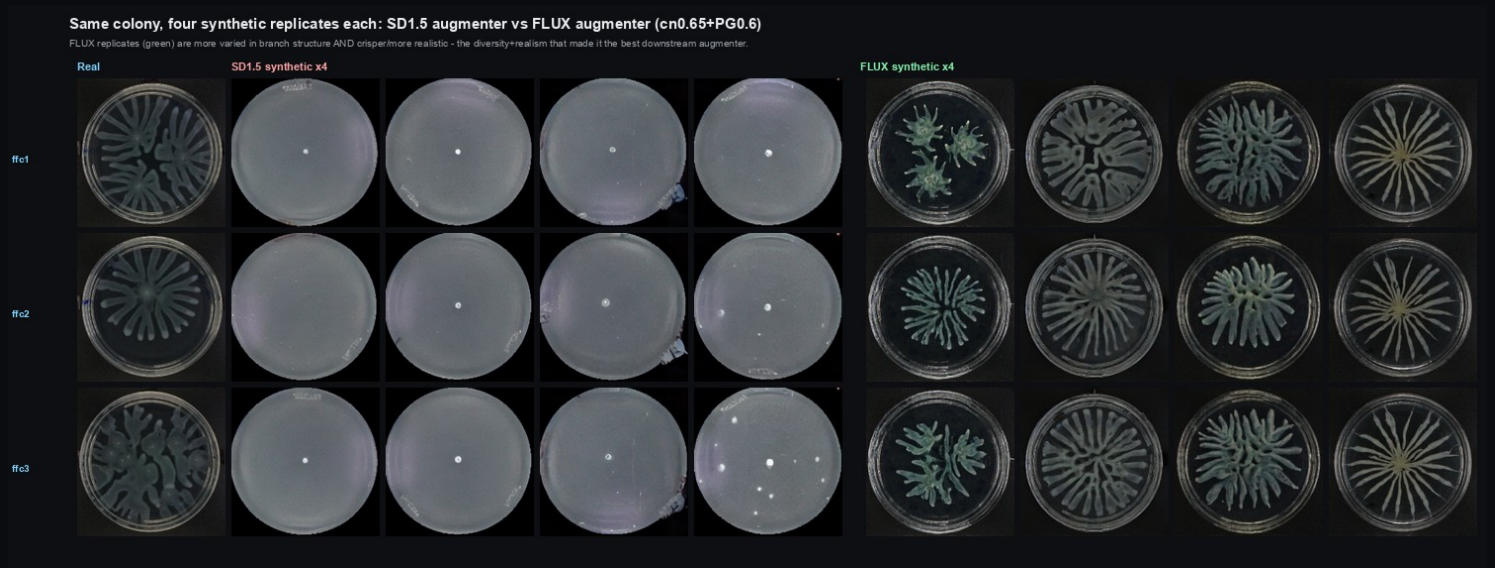
Flux backbone: beats SD1.5 on image quality AND the downstream task

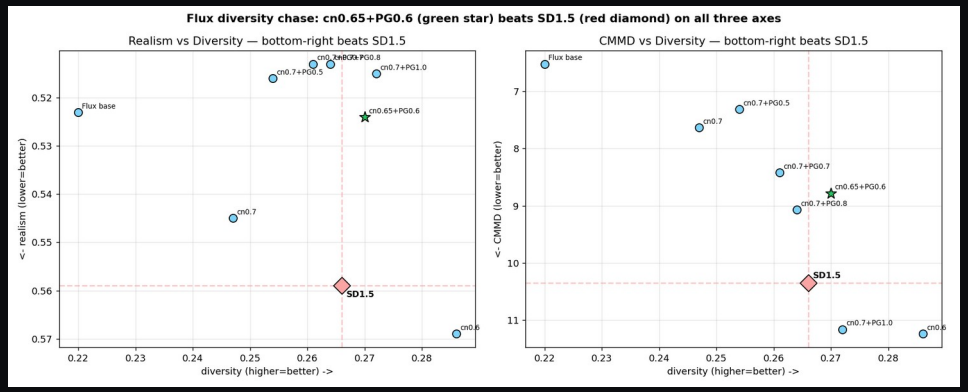
Sarthak · 2026-10-01

Key result: a fp8 Flux ControlNet augmenter (cn0.65+PG0.6) now beats-or-matches production SD1.5 on realism, CMMD, AND diversity — and is the best augmenter on the downstream test.

① Same colony, 4 synthetic replicates each — SD1.5 (washed, alike) vs FLUX (crisp, varied)

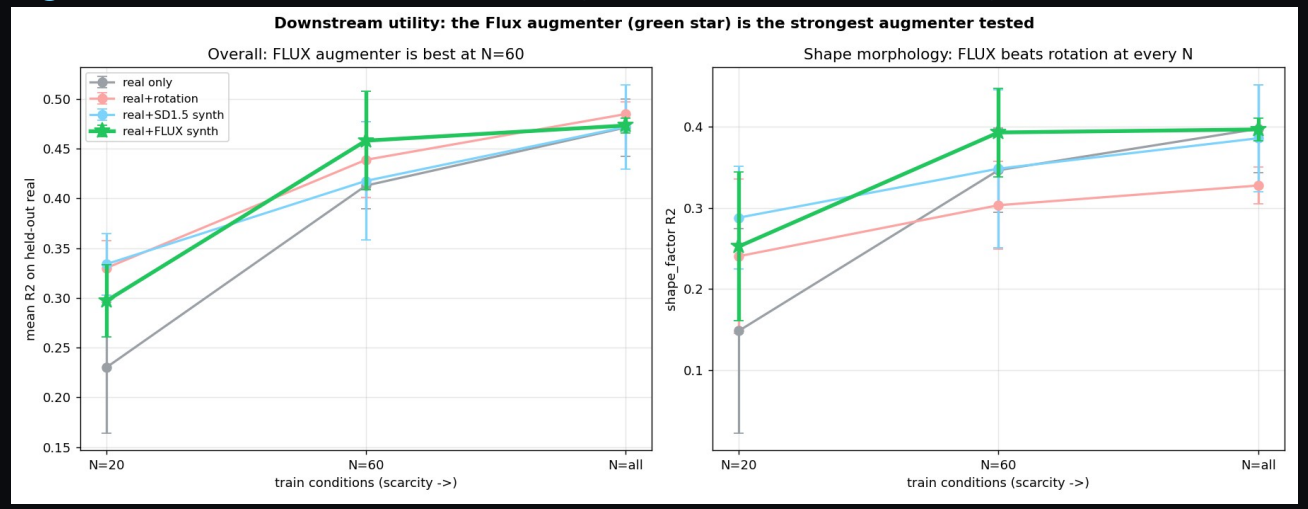


② FLUX beats SD1.5 on all 3 image metrics



realism 0.559→**0.524** CMMD ~8.8→**8.8** div 0.266→**0.270**
= Pareto-dominates SD1.5 (no worse on any axis, clearly better on realism).
Diversity lifted 0.22→0.27, inference-only.

③ DOWNSTREAM (predict colony morphology under scarcity): FLUX = best augmenter tested



④ Flux: diverse replicates / vs real



FLUX best arm @ N=60: mean R^2 **0.458** > rotation 0.439 > SD1.5-synth 0.418 > real 0.413.
Beats rotation on shape-factor (your metric) at every scarcity level.

Honest scope: one domain (branching), minimal 256px/2-block config (headroom); downstream gain real but modest.